Activities Add Instance Apr 7 5:21 PM

Virtuoso (R) Visualization & Analysis XL

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cādence

family

Subwindow: 2 Data Point Classic

TANISH_KALKAL cmos_inverter_sim...

Transient Response Tue Apr 7 17:13:31 1

Name Vis

/vin

/vout

/I0/vdd

V (V)

I (uA)

time (ns)

DC Response Tue Apr 7 17:13:31 2

Name Vis

/vin

/vout

/I0/vdd

V (V)

I (uA)

dc (V)

mouse L: M: R:

4(7) Trace: /vout; Context: /home/vlsi/simulation/cmos_inverter_simu/spectre/schematic/psf; Dataset: dc-dc

Activities Add Instance Apr 7 5:26 PM

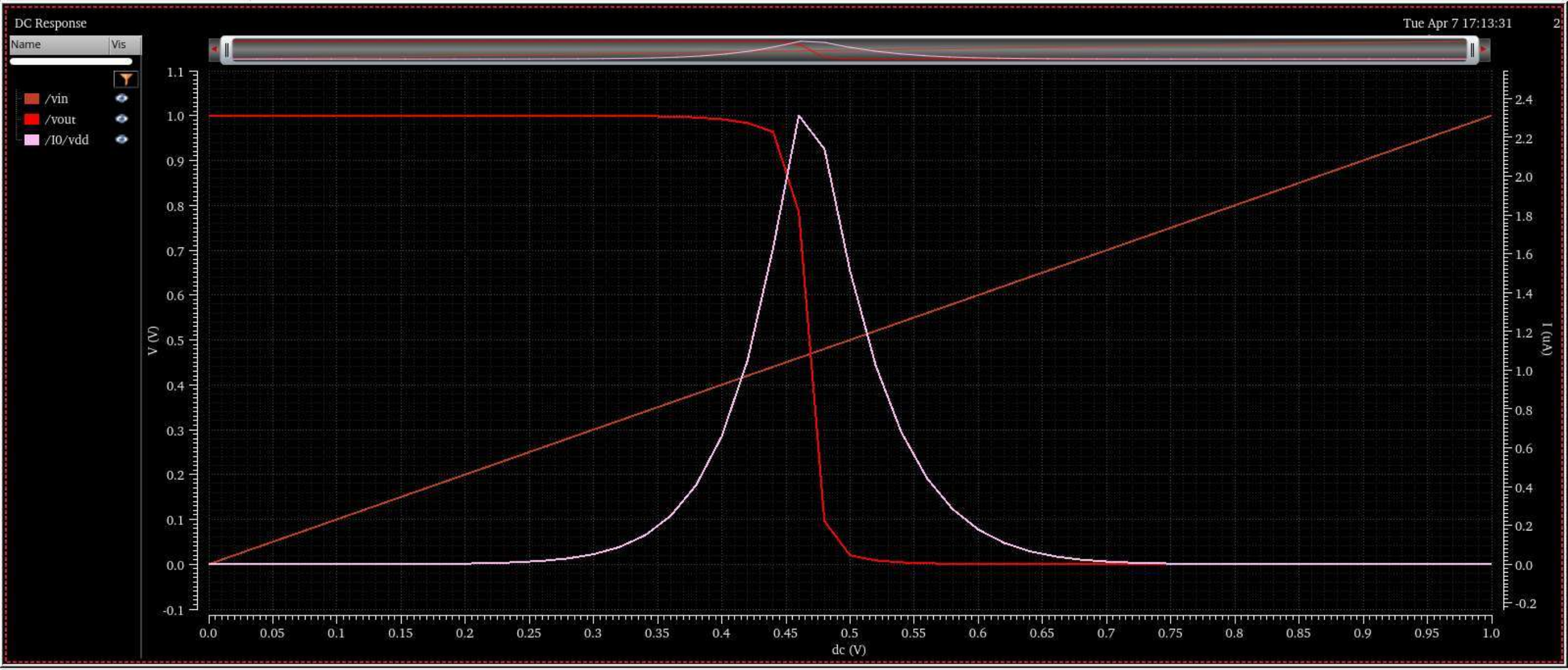
Virtuoso (R) Visualization & Analysis XL

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help

cādence

Subwindow: 2 Data Point 10.0734586731ns Classic

TANISH_KALKAL cmos_inverter_sim...



mouse L: M: R:

4(7) Trace: /vout; Context: /home/vlsi/simulation/cmos_inverter_simu/spectre/schematic/psf; Dataset: dc-dc

Activities Add Instance Apr 7 5:26 PM

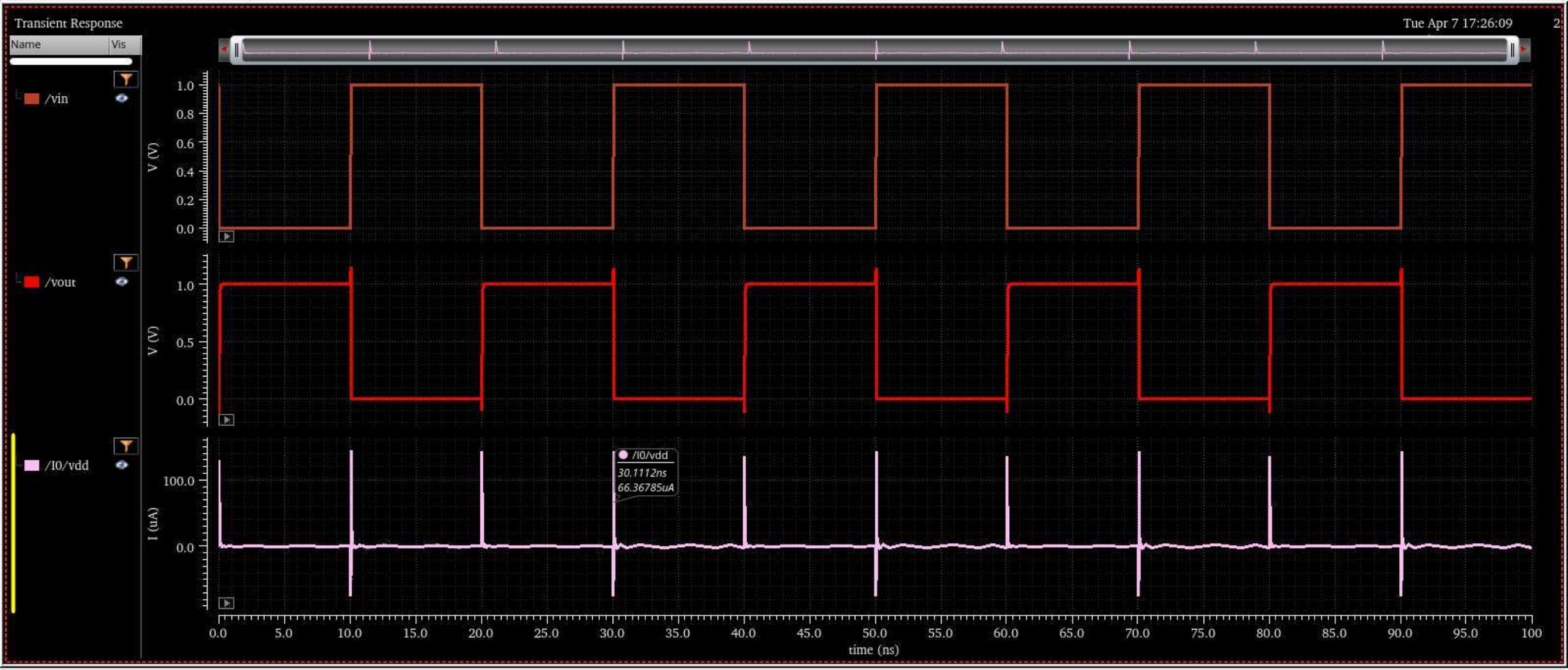
Virtuoso (R) Visualization & Analysis XL

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help cadence

family

Subwindow: 2 Data Point 10.0734586731ns Classic

TANISH_KALKAL cmos_inverter_sim...



mouse L: M: R:

4(7) Trace: /I0/vdd; Context: /home/vlsi/simulation/cmos_inverter_simu/spectre/schematic/psf; Dataset: tran-tran

Activities Virtuoso® Layout Suite XL Editing: Tanish_Nikhil_90 cmosinverter layout Apr 10 1:29 PM

Virtuoso® Layout Suite XL Editing: Tanish_Nikhil_90 cmosinverter layout

Launch File Edit View Create Verify Connectivity Options Tools Window Assura Quantus Floorplan Place Route PVS Help cadence

(F)Select:0 Sel(N):0 Sel(I):0 Sel(O):0 X 3.8700 Y 1.4800 dX 1.1200 dY -1.5200 Dist:1.8881 Cmd:

Palette

Layers

Valid Used Routing

Filter

Metal1 drawing

AV NV AS NS

gpdk090

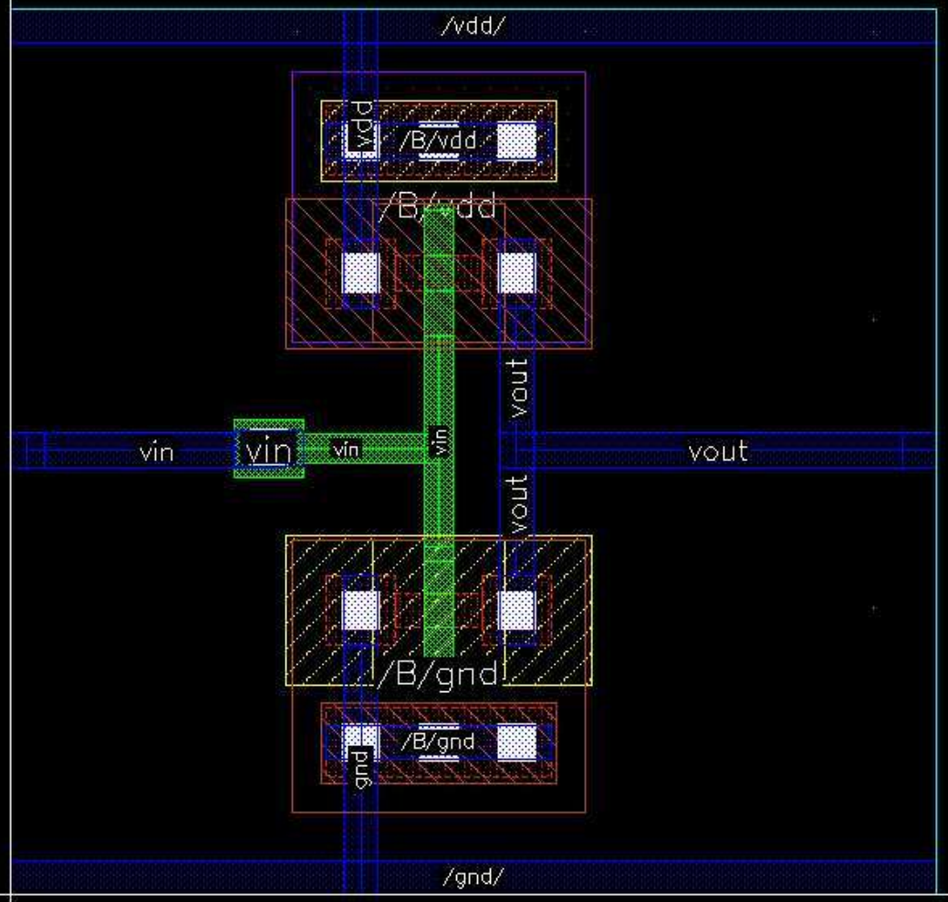
Name	Vis	Sel
All Layers		

Layer	Pu...	V	S
Poly	drw	✓	✓
Pimp	drw	✓	✓
Nhvt	drw	✓	✓
Nimp	drw	✓	✓
Phvt	drw	✓	✓
Nzvt	drw	✓	✓
SiProt	drw	✓	✓
Cont	drw	✓	✓
Metal1	drw	✓	✓
Via1	drw	✓	✓
Metal2	drw	✓	✓
Via2	drw	✓	✓
Metal3	drw	✓	✓
Via3	drw	✓	✓
Metal4	drw	✓	✓
Via4	drw	✓	✓

Objects

Objects	V	S
Instances	✓	✓
Pins	✓	✓
Vias	✓	✓

Objects Grids



mouse L: mouseSingleSelectPt() M: leHiSaveACopy(hiGetCurrentWindow()) R: _lxHiMousePopUp()

3(4) > Cmd:

Activities Add Instance Apr 11 10:33 AM

Virtuoso® Layout Suite XL Editing: Tanish_Nikhil_90 cmosinverter layout

Launch File Edit View Create Verify Connectivity Options Tools Window Assura Quantus Floorplan Place Route PVS Help cadence

Classic Off

(F)Select:0 Sel(N):0 Sel(I):0 Sel(O):0 X 4.0000 Y 0.4000 dX dY Dist: Cmd:

Palette

Layers

Valid Used Routing

Filter Filter

Nwell drawing

AV NV AS NS

gpdk090

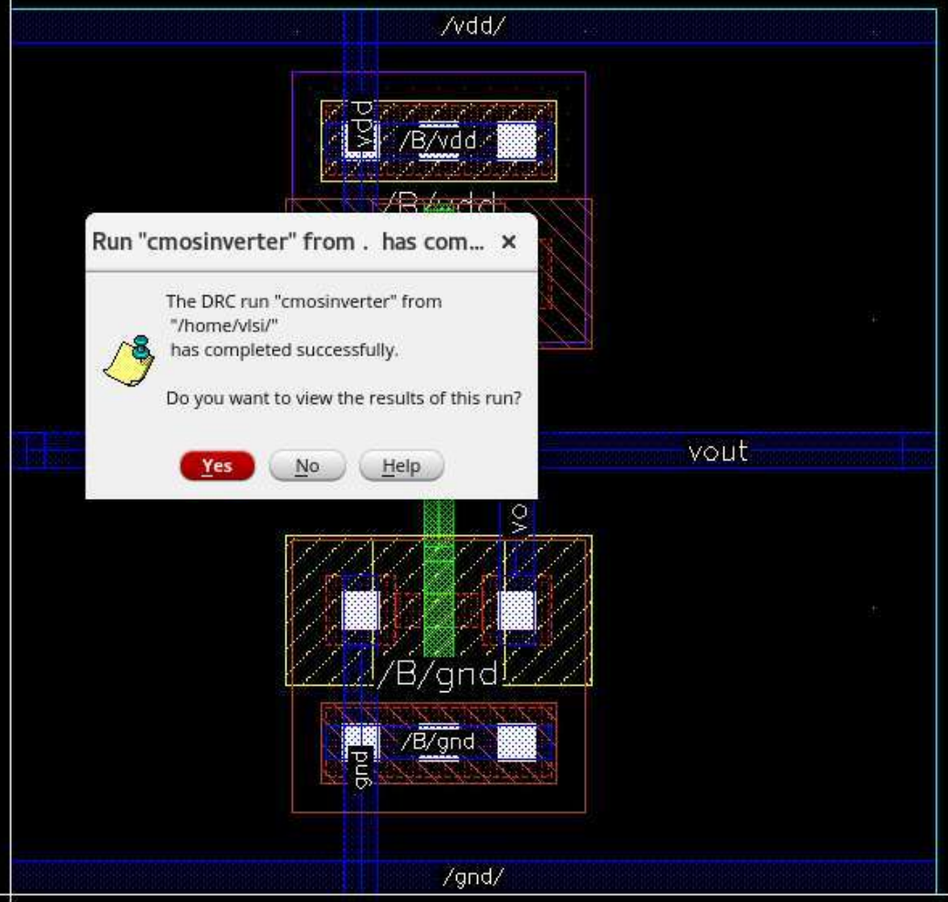
Name	Vis	Sel
All Layers		

Layer	Pu...	V	S
Nwell	drw	✓	✓
Oxide	drw	✓	✓
Oxide_thk	drw	✓	✓
Poly	drw	✓	✓
Pimp	drw	✓	✓
Nhvt	drw	✓	✓
Nimp	drw	✓	✓
Phvt	drw	✓	✓
Nzvt	drw	✓	✓
SiProt	drw	✓	✓
Cont	drw	✓	✓
Metal1	drw	✓	✓
Via1	drw	✓	✓
Metal2	drw	✓	✓
Via2	drw	✓	✓
Metal3	drw	✓	✓

Objects

Objects	V	S
Instances	✓	✓
Pins	✓	✓
Vias	✓	✓

Objects Grids



Run "cmosinverter" from . has com...

The DRC run "cmosinverter" from "/home/vlsi/" has completed successfully.

Do you want to view the results of this run?

Yes No Help

mouse L: mouseSingleSelectPt() M: vuiDRCRun() R: _lxHiMousePopUp()

7(10) > Cmd:

Activities Run Assura LVS Apr 11 10:34 AM

Virtuoso® Layout Suite XL Editing: Tanish_Nikhil_90 cmosinverter layout

Launch File Edit View Create Verify Connectivity Options Tools Window Assura Quantus Floorplan Place Route PVS Help cadence

Classic Off

(F)Select:0 Sel(N):0 Sel(I):0 Sel(O):0 X 3.7000 Y 0.6000 dX dY Dist: Cmd:

Palette

Layers

Valid Used Routing

Filter Filter

Nwell drawing

AV NV AS NS

gpdk090

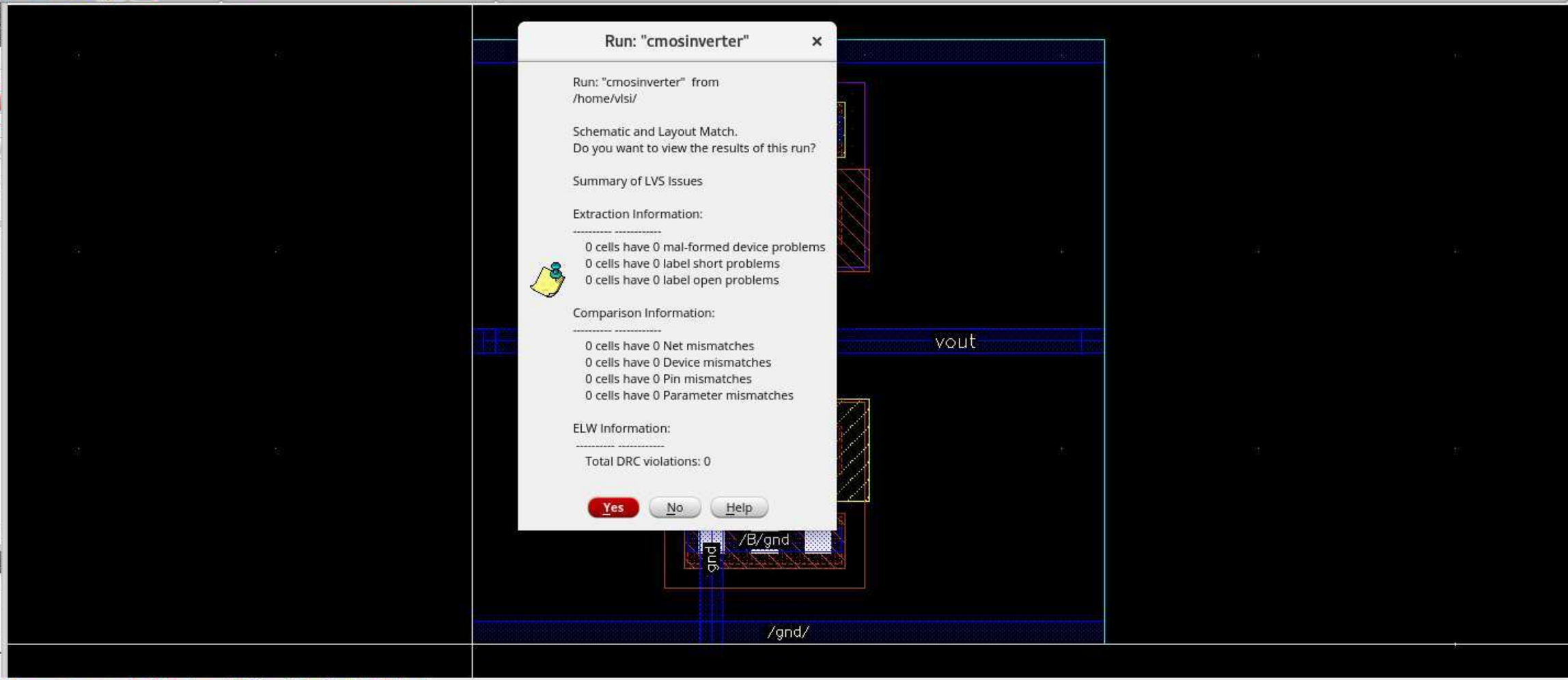
Name	Vis	Sel
All Layers	+ -	+ -

Layer	Pu...	V	S
Nwell	drw	✓	✓
Oxide	drw	✓	✓
Oxide_thk	drw	✓	✓
Poly	drw	✓	✓
Pimp	drw	✓	✓
Nhvt	drw	✓	✓
Nimp	drw	✓	✓
Phvt	drw	✓	✓
Nzvt	drw	✓	✓
SiProt	drw	✓	✓
Cont	drw	✓	✓
Metal1	drw	✓	✓
Via1	drw	✓	✓
Metal2	drw	✓	✓
Via2	drw	✓	✓
Metal3	drw	✓	✓

Objects

Objects	V	S
Instances	✓	✓
Pins	✓	✓
Vias	✓	✓

Objects Grids



Run: "cmosinverter"

Run: "cmosinverter" from
/home/vlsi/

Schematic and Layout Match.
Do you want to view the results of this run?

Summary of LVS Issues

Extraction Information:
--- ---
0 cells have 0 mal-formed device problems
0 cells have 0 label short problems
0 cells have 0 label open problems

Comparison Information:
--- ---
0 cells have 0 Net mismatches
0 cells have 0 Device mismatches
0 cells have 0 Pin mismatches
0 cells have 0 Parameter mismatches

ELW Information:
--- ---
Total DRC violations: 0

Yes No Help

mouse L: mouseSingleSelectPt() M: vuiLVSRun() R: _lxHiMousePopUp()

7(10) > Cmd:

Activities | Run Assura LVS | Apr 11 10:35 AM

Virtuoso® Layout Suite XL Editing: Tanish_Nikhil_90 cmosinverter layout

Launch File Edit View Create Verify Connectivity Options Tools Window Assura Quantus Floorplan Place Route PVS Help

cādence

Classic | Off

(F)Select:0 Sel(N):0 Sel(I):0 Sel(O):0 X -0.9000 Y 2.3000 dX dY Dist: Cmd:

Palette

Layers

LVS Debug - cmosinverter

File View Options Tools Help

cādence

Extract Compare

Cell List (sch || lay)

Summary (sch || lay)

```
*** Schematic and Layout Match
```

Open Schematic Cell... | Open Layout Cell... | Open Tool...

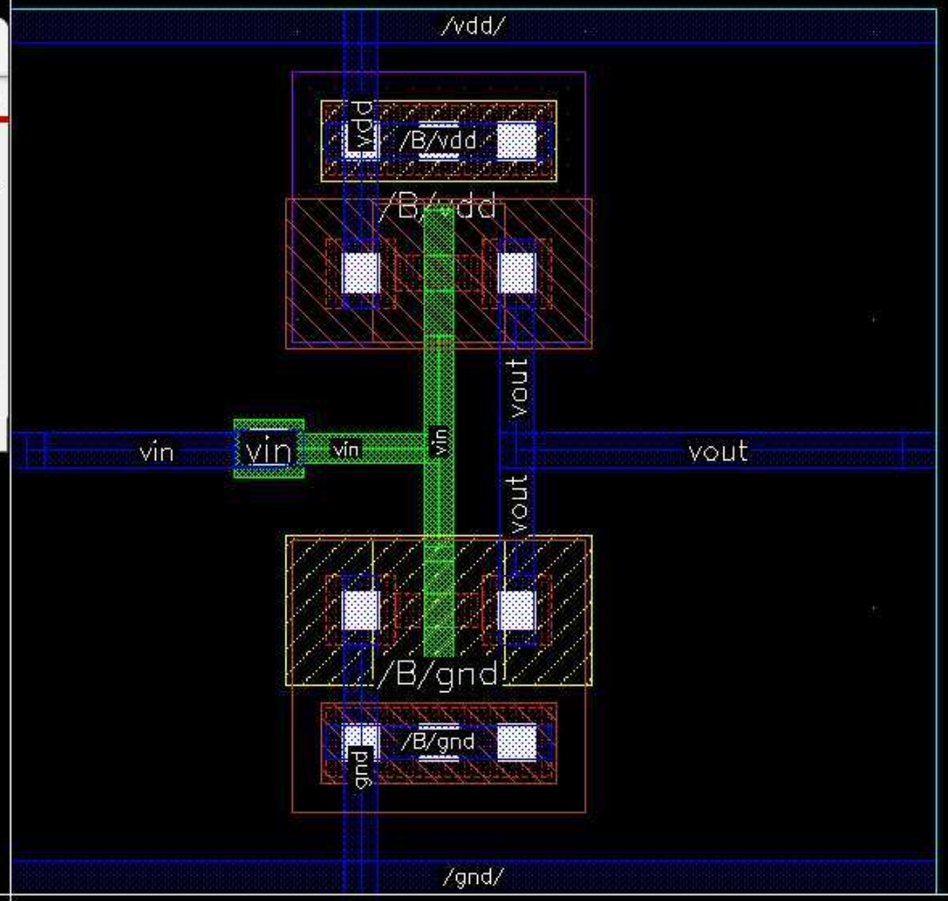
11

Nhvt	drw	✓	✓
Nimp	drw	✓	✓
Phvt	drw	✓	✓
Nzvt	drw	✓	✓
SiProt	drw	✓	✓
Cont	drw	✓	✓
Metal1	drw	✓	✓
Via1	drw	✓	✓
Metal2	drw	✓	✓
Via2	drw	✓	✓
Metal3	drw	✓	✓

Objects

Objects	V	S
Instances	✓	✓
Pins	✓	✓
Vias	✓	✓

Objects Grids



VLW

Commands

cādence

Layer Visibility

Original	AV	NV
Connect	AV	NV
Other	AV	NV

2	Metal1_p
0	Metal1_p_pintext
1	Metal1_v
1	Oxide
0	_nmos_12
0	_nmos_12_MOS_1
0	_pmos_12
0	_pmos_12_MOS_7
1	cont_ndiff
1	cont_pdiff
1	cont_poly
0	metal1_conn
0	ndiff_conn

mouse L: mouseSingleSelectPt() M: vuiLVSRun() R: _lxHiMousePopUp()

7(10) > Cmd:

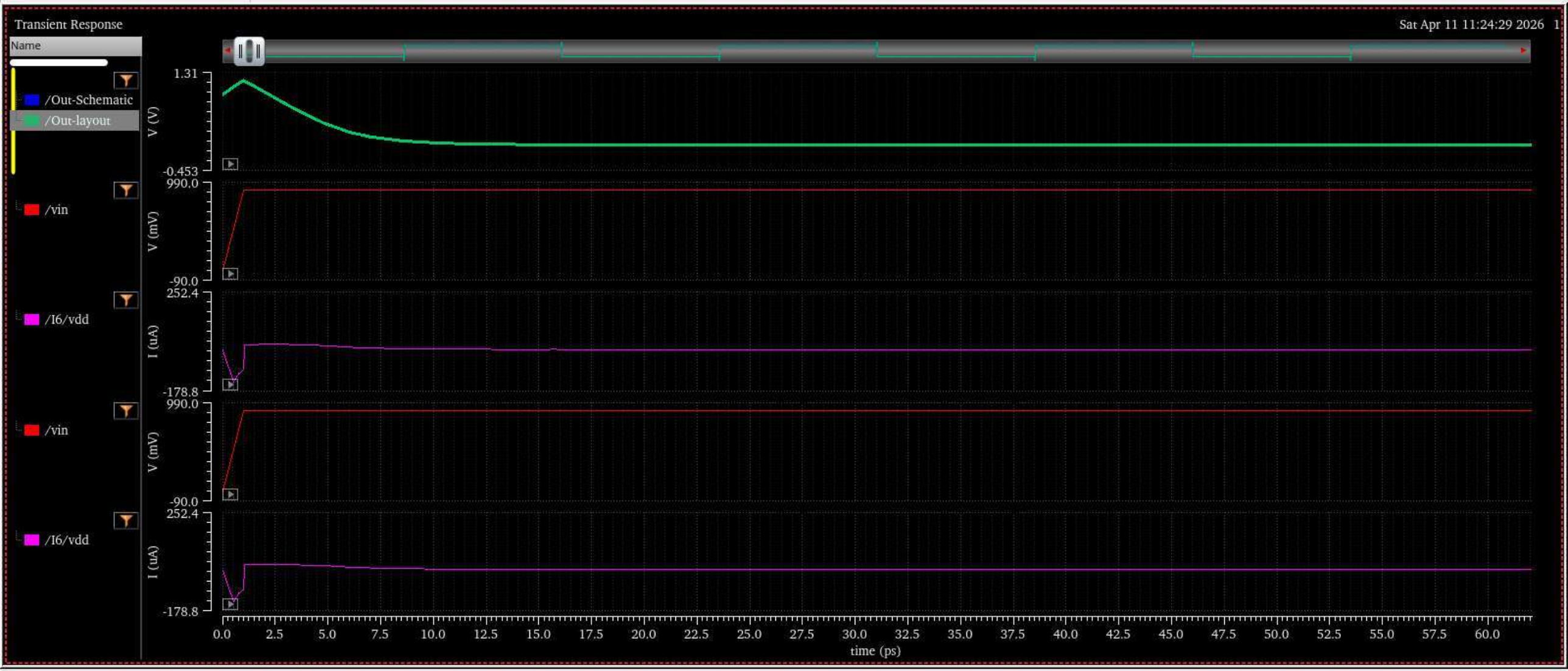
Activities Run Assura LVS Apr 11 11:57 AM

Virtuoso (R) Visualization & Analysis XL

File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help cadence

Subwindow: 1 Data Point 5.1544633542ns Classic

Tanish_Nikhil_90 CMOS_SIMULATIO...



mouse L: M: R:

13(20) Trace: /vin; Context: /home/vlsi/simulation/CMOS_SIMULATION/spectre/config/psf; Dataset: tran-tran

Activities Run Assura LVS Apr 11 11:57 AM

Virtuoso® Analog Design Environment L Editing: Tanish_Nikhil_90 CMOS_SIMULATION schematic Config: Tanish_Nikhil_90 CMOS_SIMULATION config

Launch File Edit View Create Check Options Window PVS Help cādence

ADE L Search Sim Time

Navigator

Schematic
CMOS_SIMULATION

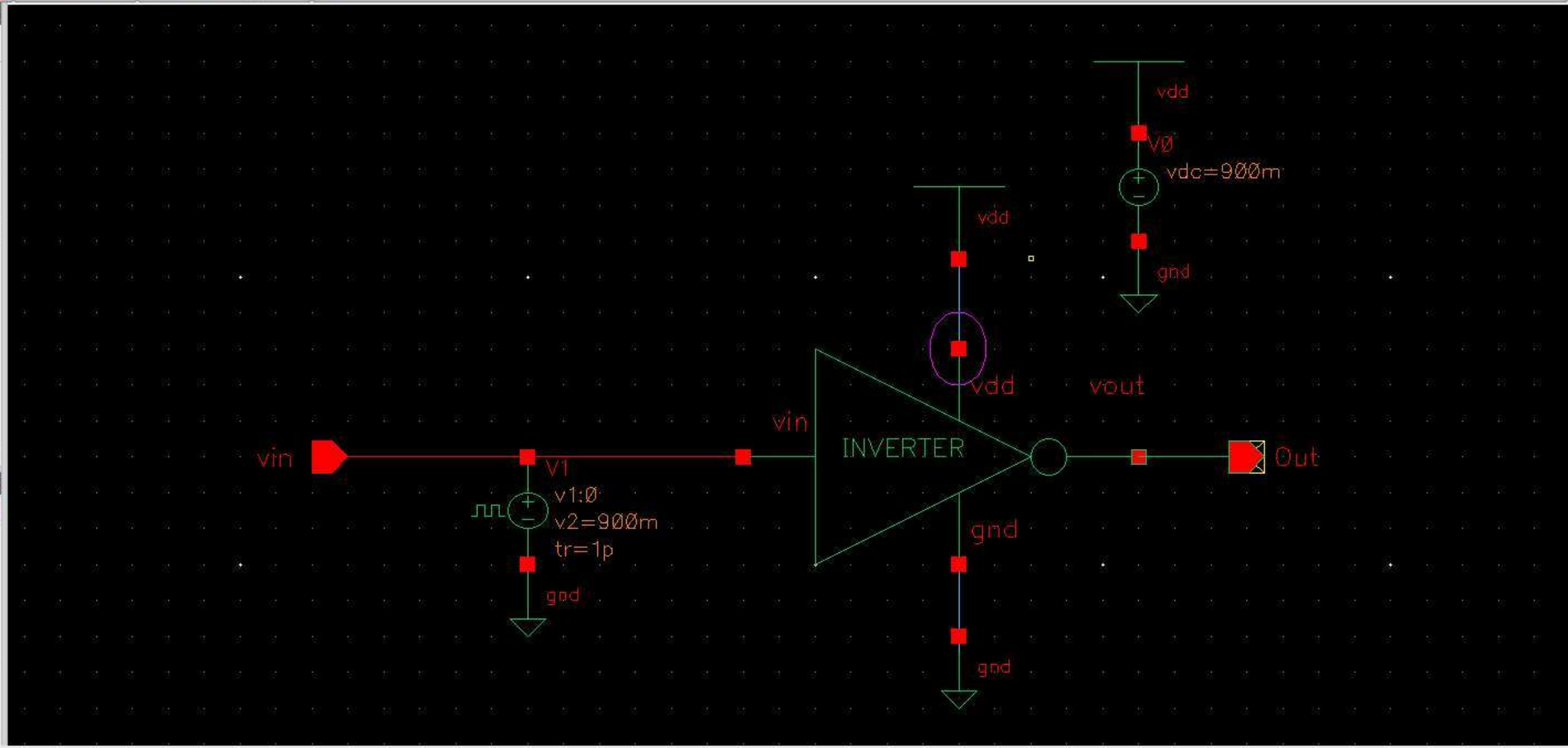
OBJECTS
- All
- Instances 8
- Nets 4
- Pins 2
- Nets and Pins

GROUPS
- Cells
- Types

Property Editor

Schemati Attributes

Library	Tanish_Nik...
Cell	CMOS_SIM...
View	schematic
Mode	editable
Last Saved	Sat Apr 11 ...
Locked By	vlsi@A17
Units	inch



mouse L: schSingleSelectPt() M: schZoomFit(1.0 0.9) R: schHiMousePopUp()

11(18) Cmd: Sel: 0 Status: Ready T=27 C Simulator: spectre aps